Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology

Department of Electrical & Computer Engineering

Course Title

Digital Techniques Sessional

Course No: ECE 2112

Lab No: 01

Report Title:

Implementation of Logic Gates Using Universal Gates, Full Adder, and Binary to BCD Converter

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Experiment 1: Implementation of OR Gate using NOR Gates

Description

In this experiment, we implement an OR gate using only NOR gates. The OR gate produces a logic HIGH (1) when at least one of its inputs is HIGH. Since the NOR gate is a universal gate, it can be used to realize any other logic gate, including the OR gate.

Procedure

1. Understanding the NOR Gate

A NOR gate is the complement of the OR gate.

$$Y = (A + B)'$$

The truth table is shown below.

A	B	$Y = (A + B)'$
0	0	1
0	1	0
1	0	0
1	1	0

Table 1: Truth Table of NOR Gate

2. Implementing the OR Gate using NOR Gates

The Boolean expression is

$$A + B = ((A + B)')'$$

Implementation steps:

- Connect inputs A and B to the first NOR gate.
- The first NOR gate produces $(A + B)'$.
- Connect this output to both inputs of another NOR gate.
- The second NOR gate acts as an inverter and produces the OR output.

3. Circuit Design

Circuit Diagram:

- Connect inputs A and B to the first NOR gate.
- Connect the output of the first NOR gate to both inputs of the second NOR gate.
- Observe the final output.

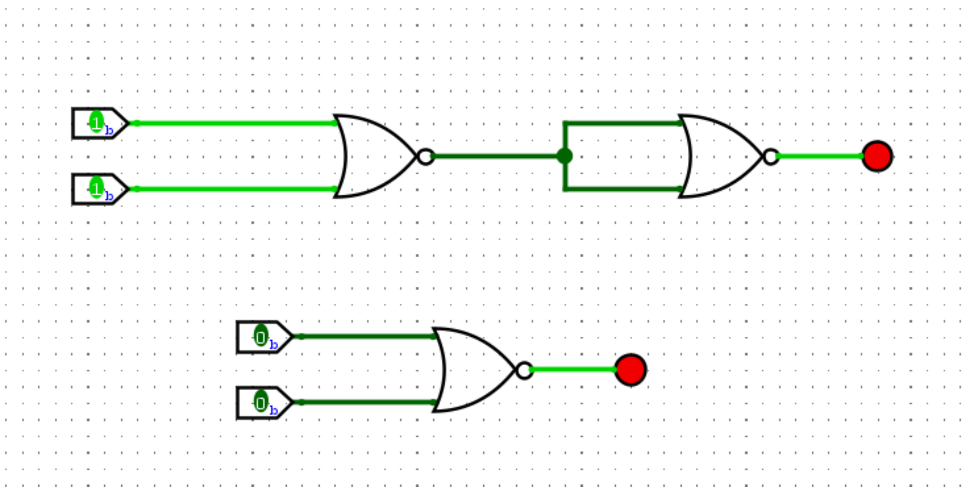

Figure 1: OR using NOR

4. Testing the Circuit

A	B	Expected Output	Obtained Output
0	0	0	0
0	1	1	1
1	0	1	1
1	1	1	1

Table 2: Testing of OR Gate using NOR Gates

The obtained outputs match the expected OR gate truth table.

Conclusion

The OR gate was successfully implemented using only NOR gates. The experimental results verified that the NOR gate is a universal gate capable of realizing an OR gate.

Experiment 2: Implementation of OR Gate using NAND Gates

Description

In this experiment, we implement an OR gate using only NAND gates. The NAND gate is a universal gate and can be used to realize an OR gate by applying De Morgan's theorem.

Procedure

1. Understanding the NAND Gate

A NAND gate is the complement of the AND gate.

$$Y = (A \cdot B)'$$

The truth table is shown below.

A	B	$Y = (A \cdot B)'$
0	0	1
0	1	1
1	0	1
1	1	0

Table 3: Truth Table of NAND Gate

2. Implementing the OR Gate using NAND Gates

The Boolean expression is

$$A + B = (A' \cdot B')'$$

Implementation steps:

- Use the first NAND gate as an inverter to obtain A'.
- Use the second NAND gate as an inverter to obtain B'.
- Connect the outputs to a third NAND gate.
- The output of the third NAND gate is $A + B$.

3. Circuit Design

Circuit Diagram:

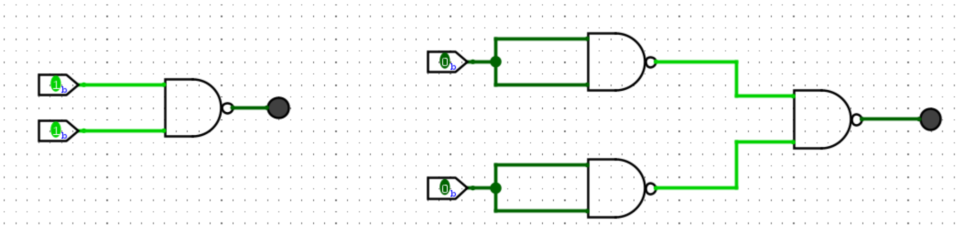

Figure 2: OR using NAND

- Connect input A to both terminals of the first NAND gate.
- Connect input B to both terminals of the second NAND gate.
- Connect the outputs of the first two NAND gates to the third NAND gate.
- Observe the final output.

4. Testing the Circuit

A	B	Expected Output	Obtained Output
0	0	0	0
0	1	1	1
1	0	1	1
1	1	1	1

Table 4: Testing of OR Gate using NAND Gates

The obtained outputs match the expected OR gate truth table.

Conclusion

The OR gate was successfully implemented using only NAND gates. The experimental results verified that the NAND gate is a universal gate capable of realizing an OR gate.

Experiment 3: Implementation of AND Gate using NOR Gates

Description

In this experiment, we implement an AND gate using only NOR gates. Since the NOR gate is a universal gate, it can be used to realize an AND gate by applying De Morgan's theorem.

Procedure

1. Understanding the AND Gate

An AND gate produces HIGH only when both inputs are HIGH.

$$Y = A \cdot B$$

The truth table is shown below.

A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

Table 5: Truth Table of AND Gate

2. Implementing the AND Gate using NOR Gates

The Boolean expression is

$$A \cdot B = (A' + B')'$$

Implementation steps:

- Use the first NOR gate as an inverter to obtain A'.
- Use the second NOR gate as an inverter to obtain B'.
- Connect the outputs to a third NOR gate.
- The output of the third NOR gate is $A \cdot B$.

3. Circuit Design

Circuit Diagram:

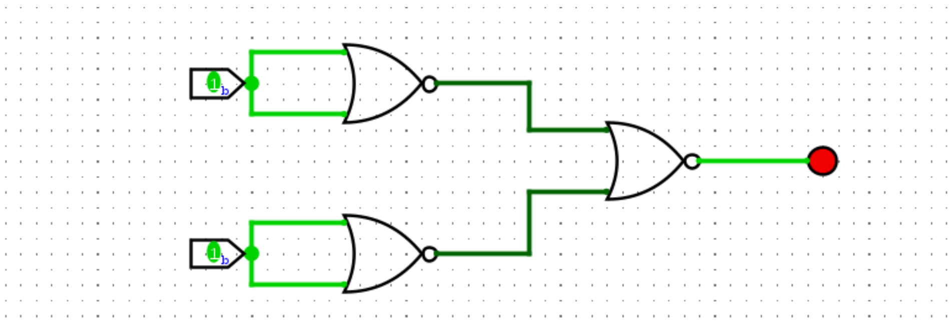

Figure 3: AND using NOR

- Connect input A to both terminals of the first NOR gate.
- Connect input B to both terminals of the second NOR gate.
- Connect the outputs of the first two NOR gates to the third NOR gate.
- Observe the final output.

4. Testing the Circuit

A	B	Expected Output	Obtained Output
0	0	0	0
0	1	0	0
1	0	0	0
1	1	1	1

Table 6: Testing of AND Gate using NOR Gates

The obtained outputs match the expected AND gate truth table.

Conclusion

The AND gate was successfully implemented using only NOR gates. The experimental results verified that the NOR gate is a universal gate capable of realizing an AND gate.

Experiment 4: Implementation of AND Gate using NAND Gates

Description

In this experiment, we implement an AND gate using only NAND gates. Since the NAND gate is a universal gate, it can be used to realize an AND gate by connecting two NAND gates appropriately.

Procedure

1. Understanding the AND Gate

An AND gate produces HIGH only when both inputs are HIGH.

$$Y = A \cdot B$$

The truth table is shown below.

A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

Table 7: Truth Table of AND Gate

2. Implementing the AND Gate using NAND Gates

The Boolean expression is

$$A \cdot B = ((A \cdot B)')'$$

Implementation steps:

- Apply inputs A and B to the first NAND gate.
- The output becomes $(A \cdot B)'$.
- Connect this output to both inputs of a second NAND gate.
- The second NAND gate inverts the signal and produces $A \cdot B$.

3. Circuit Design

Circuit Diagram:

- Connect inputs A and B to the first NAND gate.
- Connect the output of the first NAND gate to both inputs of the second NAND gate.
- Observe the final output.

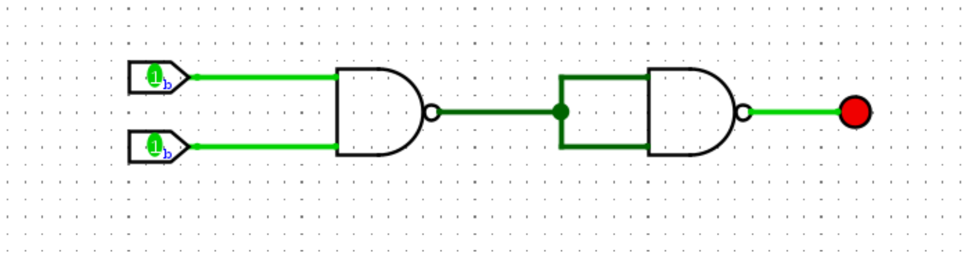

Figure 4: AND using NAND

4. Testing the Circuit

A	B	Expected Output	Obtained Output
0	0	0	0
0	1	0	0
1	0	0	0
1	1	1	1

Table 8: Testing of AND Gate using NAND Gates

The obtained output matches the expected AND gate truth table.

Conclusion

The AND gate was successfully implemented using only NAND gates. The experimental results verified that the NAND gate is a universal gate capable of realizing an AND gate.

Experiment 5: Implementation of NOT Gate using NOR Gates

Description

In this experiment, we implement a NOT gate using a NOR gate. A NOR gate acts as an inverter when both of its inputs are connected together. This experiment demonstrates how a universal gate can be used to realize a NOT gate.

Procedure

1. Understanding the NOT Gate

A NOT gate produces the complement of the input.

$$Y = A'$$

The truth table is shown below.

A	$Y = A'$
0	1
1	0

Table 9: Truth Table of NOT Gate

2. Implementing the NOT Gate using NOR Gates

The Boolean expression is

$$A' = A \text{ NOR } A$$

Implementation steps:

- Connect input A to both inputs of the NOR gate.
- Observe the output.
- The output is the complement of the input.

3. Circuit Design

Circuit Diagram:

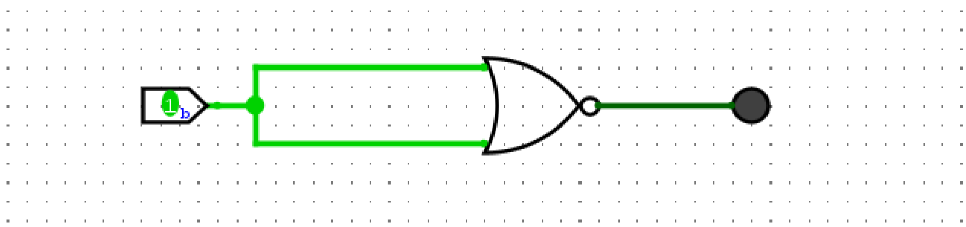

Figure 5: NOT using NOR

- Connect input A to both terminals of the NOR gate.
- Observe the output.

4. Testing the Circuit

A	Expected Output	Obtained Output
0	1	1
1	0	0

Table 10: Testing of NOT Gate using NOR Gate

The obtained outputs match the expected NOT gate truth table.

Conclusion

The NOT gate was successfully implemented using a NOR gate. The experiment confirms that a NOR gate can function as an inverter when both of its inputs are connected together.

Experiment 6: Implementation of NOT Gate using NAND Gates

Description

In this experiment, we implement a NOT gate using a NAND gate. A NAND gate is a universal gate and can perform the NOT operation by connecting both of its inputs together. This experiment demonstrates the implementation and verification of a NOT gate using only a NAND gate.

Procedure

1. Understanding the NOT Gate

A NOT gate produces the complement of the input.

$$Y = A'$$

The truth table is shown below.

A	$Y = A'$
0	1
1	0

Table 11: Truth Table of NOT Gate

2. Implementing the NOT Gate using NAND Gates

The Boolean expression is

$$A' = A \text{ NAND } A$$

Implementation steps:

- Connect input A to both inputs of the NAND gate.
- Observe the output.
- The output obtained is the complement of the input.

3. Circuit Design

Circuit Diagram:

- Connect input A to both terminals of the NAND gate.
- Observe the output.

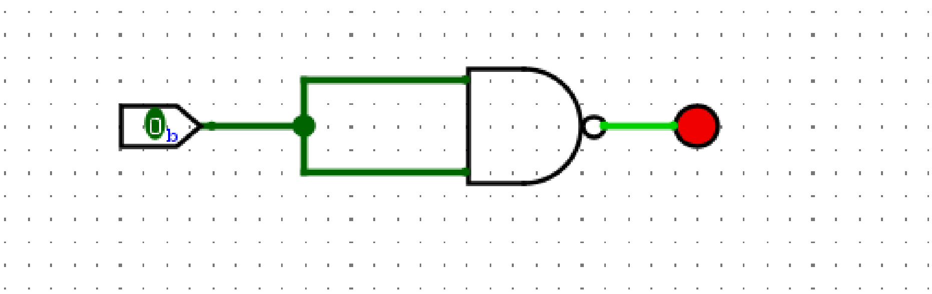

Figure 6: NOT using NAND

4. Testing the Circuit

A	Expected Output	Obtained Output
0	1	1
1	0	0

Table 12: Testing of NOT Gate using NAND Gate

The obtained outputs match the expected NOT gate truth table.

Conclusion

In this experiment, the NOT gate was successfully implemented using a NAND gate. The output obtained for each input matched the expected NOT gate truth table. This experiment confirms that the NAND gate is a universal gate and can be used as an inverter by connecting both of its inputs together.

Experiment 7: Implementation of Full Adder and Testing

Description

In this experiment, we implement a Full Adder circuit using two Half Adders and one OR gate. A Full Adder is a combinational logic circuit that adds three binary inputs (A, B, and Cin) and produces two outputs: Sum and Carry Out (Cout). This experiment verifies the operation of a Full Adder for all possible input combinations.

Procedure

1. Understanding the Full Adder

A Full Adder performs the addition of three binary inputs.
The Boolean expressions are:

$$Sum = A \oplus B \oplus Cin$$

$$Cout = AB + Cin(A \oplus B)$$

The truth table is shown below.

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Table 13: Truth Table of Full Adder

2. Implementing the Full Adder

Implementation steps:

- Construct the first Half Adder using inputs A and B.
- Connect the Sum output of the first Half Adder and the Carry input (Cin) to the second Half Adder.
- Connect the Carry outputs of both Half Adders to an OR gate.
- The output of the second Half Adder is the final Sum.
- The output of the OR gate is the final Carry Out (Cout).

3. Circuit Design

Circuit Diagram:

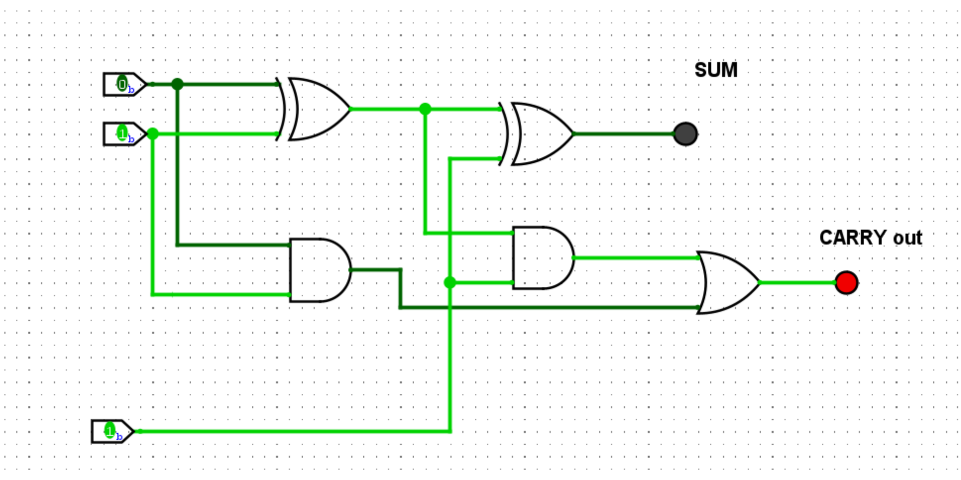

Figure 7: FULL ADDER circuit

- Connect two Half Adder circuits in cascade.
- Connect the Carry outputs to an OR gate.
- Observe the outputs Sum and Cout.

4. Testing the Circuit

Apply all eight possible input combinations.

A	B	Cin	Expected Output (Sum, Cout)	Obtained Output
0	0	0	0,0	0,0
0	0	1	1,0	1,0
0	1	0	1,0	1,0
0	1	1	0,1	0,1
1	0	0	1,0	1,0
1	0	1	0,1	0,1
1	1	0	0,1	0,1
1	1	1	1,1	1,1

Table 14: Testing of Full Adder

The obtained outputs match the expected Full Adder truth table for all input combinations.

Conclusion

In this experiment, a Full Adder circuit was successfully implemented using two Half Adders and one OR gate. The outputs obtained for all input combinations matched the theoretical truth table. This experiment demonstrates the operation of binary addition and the practical implementation of a Full Adder using basic logic gates.

Experiment 8: Implementation of Binary to BCD Converter

Description

In this experiment, we implement a Binary to BCD (Binary-Coded Decimal) Converter using basic logic gates. A Binary to BCD Converter converts a 4-bit binary number into its equivalent BCD representation. This conversion is widely used in digital systems for displaying decimal numbers on seven-segment displays.

Procedure

1. Understanding Binary to BCD Conversion

A Binary to BCD Converter accepts a 4-bit binary input and generates the corresponding BCD output.

For example,

$$1010_2 = 10_{10}$$

The BCD representation of decimal 10 is:

$$0001\ 0000$$

The truth table for a 4-bit Binary to BCD Converter is shown below.

B_3	B_2	B_1	B_0	D_4	D_3	D_2	D_1
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	0
0	0	1	1	0	0	1	1
0	1	0	0	0	1	0	0
0	1	0	1	0	1	0	1
0	1	1	0	0	1	1	0
0	1	1	1	0	1	1	1
1	0	0	0	1	0	0	0
1	0	0	1	1	0	0	1
1	0	1	0	1	0	1	0
1	0	1	1	1	0	1	1
1	1	0	0	1	1	0	0
1	1	0	1	1	1	0	1
1	1	1	0	1	1	1	0
1	1	1	1	1	1	1	1

Table 15: Truth Table of Binary to BCD Converter

2. Implementing the Binary to BCD Converter

Implementation steps:

- Connect the four binary inputs (B_3, B_2, B_1, and B_0) to the Binary to BCD Converter circuit.
- The converter generates the corresponding BCD outputs.
- Connect the BCD outputs to a BCD-to-Seven Segment Display.
- Observe the displayed decimal digit for different binary inputs.

3. Circuit Design

Circuit Diagram:

- Connect four input switches to the Binary to BCD Converter.
- Connect the output of the converter to the BCD-to-Seven Segment Display.
- Verify the displayed decimal value.

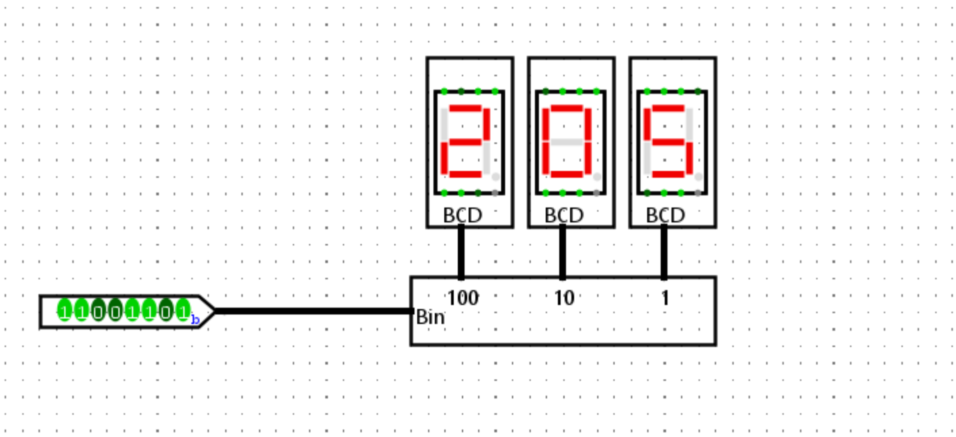

Figure 8: Binary to BCD

4. Testing the Circuit

Apply different combinations of binary inputs and observe the corresponding BCD outputs.

Binary Input	Decimal Equivalent	BCD Output
0000	0	0000
0001	1	0001
0101	5	0101
1001	9	1001
1010	10	0001 0000
1111	15	0001 0101

Table 16: Testing of Binary to BCD Converter

The observed BCD outputs matched the expected decimal values for all applied binary inputs.

Conclusion

In this experiment, a Binary to BCD Converter was successfully implemented and tested. The circuit correctly converted 4-bit binary inputs into their corresponding BCD outputs. The results verified the correct operation of the converter and demonstrated its application in digital display systems.

References

1. M. M. Mano, *Digital Design*, 5th ed. Upper Saddle River, N.J.: Pearson, 2013.

2. R. J. Tocci, N. S. Widmer, and G. L. Moss, *Digital Systems: Principles and Applications*, 11th ed. Upper Saddle River, N.J.: Pearson, 2011.